Chapter 9, Problem 33P

Problem

A series *RL* circuit is connected to a 220-V ac source. If the voltage across the resistor is 170 V, find the voltage across the inductor.

Step-by-step solution

Step 1 of 2

The source voltage in a series *RL* circuit can be expressed as the sum of the magnitudes of voltage across resistor and inductor, mathematically it can be written as follows:

$$|\mathbf{V_S}| = |\mathbf{V_R}| + |\mathbf{V_L}|$$
$$\mathbf{V_S^2} = \mathbf{V_R^2} + \mathbf{V_L^2}$$
$$\mathbf{V_L^2} = \mathbf{V_S^2} - \mathbf{V_R^2}$$
$$\mathbf{V_L} = \sqrt{\mathbf{V_S^2} - \mathbf{V_R^2}}$$

Step 2 of 2

Substitute 220 V for $\mathbf{V_S}$ and 170 V for $\mathbf{V_R}$.

$$\begin{aligned}\mathbf{V_L} &= \sqrt{\mathbf{V_S^2} - \mathbf{V_R^2}}\\ &= \sqrt{(220)^2 - (170)^2}\\ &= 139.64\text{ V}\end{aligned}$$

Therefore, the voltage across the inductor is $\boxed{139.64\text{ V}}$.